

Complex Potential in 2D Fluid Flow

MA 4410 – Complex Analysis

Motivation

- ▶ Many 2D incompressible, irrotational flows admit a complex analytic description.
- ▶ This allows us to use tools from complex analysis to solve fluid problems.
- ▶ Key idea: encode velocity and geometry into a single complex function.

Velocity Field in 2D

Let the velocity field be

$$\mathbf{u}(x, y) = (u(x, y), v(x, y)).$$

► Incompressibility:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

► Irrotationality:

$$\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0$$

These are exactly the Cauchy–Riemann equations (up to sign conventions).

Velocity Potential and Stream Function

- ▶ Define the velocity potential $\phi(x, y)$:

$$\nabla\phi = (u, v)$$

- ▶ Define the stream function $\psi(x, y)$:

$$\nabla^\perp\psi = (u, v), \quad \left(\frac{\partial\psi}{\partial y}, -\frac{\partial\psi}{\partial x}\right) = (u, v)$$

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Cauchy–Riemann relations:

$$\frac{\partial\phi}{\partial x} = \frac{\partial\psi}{\partial y}, \quad \frac{\partial\phi}{\partial y} = -\frac{\partial\psi}{\partial x}$$

Complex Potential

Define the complex potential:

$$w(z) = \phi(x, y) + i\psi(x, y), \quad z = x + iy.$$

- ▶ w is analytic $\iff$ flow is incompressible and irrotational.
- ▶ Equipotential curves: $\phi = \text{const}$
- ▶ Streamlines: $\psi = \text{const}$
- ▶ These curves are orthogonal.

Complex Velocity

The complex velocity is

$$\frac{dw}{dz} = u - iv.$$

- Magnitude of velocity:

$$|\mathbf{u}| = \left| \frac{dw}{dz} \right|$$

- Direction encoded in argument of $w'(z)$.

Laplace's Equation

Both ϕ and ψ are harmonic:

$$\Delta\phi = 0, \quad \Delta\psi = 0.$$

- ▶ Follows from analyticity of $w(z)$.
- ▶ Thus fluid problems reduce to solving Laplace's equation.

Basic Example: Uniform Flow

$$w(z) = Uz$$

- ▶ $U \in \mathbb{R}$ is constant velocity.
- ▶ $\phi = Ux, \quad \psi = Uy$
- ▶ Streamlines are horizontal lines.

Source / Sink

$$w(z) = \frac{Q}{2\pi} \log z$$

- ▶ Radial flow outward (source) or inward (sink).
- ▶ $\psi = \frac{Q}{2\pi} \arg z$
- ▶ Streamlines are radial lines.
- ▶ Equipotentials are circles.

Point Vortex

$$w(z) = -\frac{i\Gamma}{2\pi} \log z$$

- ▶ Circular flow around origin.
- ▶ $\phi = \frac{\Gamma}{2\pi} \arg z$
- ▶ $\psi = -\frac{\Gamma}{2\pi} \log |z|$

Doublet (Dipole)

$$w(z) = \frac{\kappa}{z}$$

- ▶ Models flow around small obstacle.
- ▶ Can be obtained as limit of source-sink pair.

Superposition Principle

- ▶ Linear combination of solutions is again a solution:

$$w(z) = w_1(z) + w_2(z)$$

- ▶ Allows construction of complex flows from simple pieces.

Flow Around a Cylinder

$$w(z) = Uz + \frac{\kappa}{z}$$

- ▶ Combination of uniform flow and doublet.
- ▶ Produces flow past a circular cylinder.
- ▶ Stagnation points occur where $w'(z) = 0$.

Conformal Mapping

- ▶ If $w(z)$ describes a flow, and $z = f(\zeta)$ is analytic,

$$W(\zeta) = w(f(\zeta))$$

also describes a flow.

- ▶ Used to transform simple geometries into complex ones.

Joukowski Transform

The Joukowski map is defined by

$$\zeta = z + \frac{c^2}{z}, \quad c > 0.$$

- ▶ Maps circles in the z -plane to airfoil-like shapes in the ζ -plane.
- ▶ Singular at $z = 0$.
- ▶ Key idea: use conformal mapping to transform known flows.

Geometry of the Mapping

Let a circle in the z -plane be

$$z = z_0 + Re^{i\theta}.$$

- ▶ If the circle encloses the origin, the image is an airfoil.
- ▶ If the circle passes through $z = c$, a cusp forms in the image.
- ▶ This cusp corresponds to the trailing edge.

Critical Points and Cusp Formation

Compute:

$$\frac{d\zeta}{dz} = 1 - \frac{c^2}{z^2}.$$

- ▶ Critical points occur when

$$z^2 = c^2 \quad \Rightarrow \quad z = \pm c.$$

- ▶ If the circle passes through $z = c$, then $\zeta'(z) = 0$ there.
- ▶ This produces a cusp in the mapped boundary.

Flow Around a Circle

Start with the classical potential:

$$w(z) = Uz + \frac{\kappa}{z} - \frac{i\Gamma}{2\pi} \log z.$$

- ▶ Uniform flow: Uz
- ▶ Doublet: κ/z
- ▶ Circulation: $-\frac{i\Gamma}{2\pi} \log z$

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$$\text{Velocity: } w'(z) = U - \frac{\kappa}{z^2} - \frac{i\Gamma}{2\pi z}.$$

Cylinder Condition

To model flow around a cylinder of radius R , choose

$$\kappa = UR^2.$$

- ▶ Then $|z| = R$ is a streamline.
- ▶ This gives flow past a circular obstacle.

Mapping to an Airfoil

Now apply the Joukowski transform:

$$\zeta = z + \frac{c^2}{z}.$$

- ▶ The circle $|z - z_0| = R$ maps to an airfoil.
- ▶ The complex potential in the ζ -plane is

$$W(\zeta) = w(z(\zeta)).$$

Velocity in the Physical Plane

Using the chain rule:

$$\frac{dW}{d\zeta} = \frac{dw}{dz} \cdot \frac{dz}{d\zeta}.$$

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Since

$$\frac{d\zeta}{dz} = 1 - \frac{c^2}{z^2}, \quad \frac{dz}{d\zeta} = \left(1 - \frac{c^2}{z^2}\right)^{-1},$$

we get:

$$\frac{dW}{d\zeta} = \frac{w'(z)}{1 - \frac{c^2}{z^2}}.$$

Trailing Edge and the Kutta Condition

- ▶ At the trailing edge: $z = c$, so

$$\frac{d\zeta}{dz} = 0.$$

- ▶ This can cause velocity blow-up unless numerator also vanishes.

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Kutta condition:

$$w'(z) = 0 \quad \text{at } z = c.$$

- ▶ Determines the circulation Γ .
- ▶ Ensures finite velocity at the trailing edge.

Solving for Circulation

Impose:

$$U - \frac{\kappa}{c^2} - \frac{i\Gamma}{2\pi c} = 0.$$

- ▶ Using $\kappa = UR^2$ gives an explicit formula for Γ .
- ▶ This produces lift on the airfoil.

Physical Interpretation

- ▶ Mapping turns a simple cylinder flow into airfoil flow.
- ▶ Circulation Γ produces lift.
- ▶ Geometry controlled by:
 - ▶ c (mapping parameter)
 - ▶ circle center z_0
 - ▶ radius R

Summary of the Construction

1. Start with flow around a circle in z -plane.
2. Choose parameters so boundary is a streamline.
3. Apply

$$\zeta = z + \frac{c^2}{z}.$$

4. Impose Kutta condition at $z = c$.
5. Obtain physically realistic airfoil flow.

Physical Interpretation

- ▶ ϕ : potential energy / Bernoulli function (in steady flow)
- ▶ ψ : stream function (flow lines)
- ▶ Orthogonality: equipotentials $\perp$ streamlines

Limitations

- ▶ Assumes:
 - ▶ Inviscid flow
 - ▶ Irrotational flow (except at singularities)
 - ▶ 2D geometry
- ▶ Does not capture boundary layers or turbulence.

Summary

- ▶ Complex potential $w(z) = \phi + i\psi$ encodes fluid flow.
- ▶ Analyticity $\iff$ incompressible + irrotational.
- ▶ Many flows constructed via superposition and conformal maps.
- ▶ Powerful bridge between complex analysis and physics.